

Antimicrobial and antioxidant coating based on basil seed gum incorporated with Shirazi thyme and summer savory essential oils emulsions for shelf-life extension of refrigerated chicken fillets

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ABSTRACT

The present study was conducted to evaluate the effects of antimicrobial and antioxidant coatings based on basil-seed gum (BSG) on the refrigerated shelf-life of chicken fillets. Antimicrobial and antioxidant properties were developed by incorporating summer savory essential oil (SEO) and Shirazi thyme essential oil (TEO) emulsions, individually or in combination with each other, into BSG coating. Microbiological evaluations including total viable count (TVC), psychrotrophic (PBC) and lactic acid bacteria counts (LAB), as well as chemical characteristics namely peroxide value (PV), thiobarbituric acid (TBA), total volatile basic nitrogen (TVB-N) and pH were performed on coated and control samples. Moreover, sensory evaluations were performed for odor, color, texture and overall acceptability. Results showed that incorporating SEO and TEO into BSG coatings could significantly ($p < 0.5$) increase the quality of chicken fillets. Antimicrobial and antioxidant coatings significantly ($p < 0.05$) reduced pH, TVB-N, PV and TBA values compared to uncoated control samples, specially for BSG/TEO coated samples, which resulted the best treatment. pH, TVB-N, PV and TBA in BSG/TEO coated samples were 5.96, 16.84, 3.33 and 2.44, respectively, lower than those of uncoated control samples on day 12 of storage. A similar reduction trend was observed for microbial population. TVC, PBC and LAB were 3.69, 4.73, and 3.87 log CFU/g on day 12 of storage in BSG/TEO coated samples lower than those of uncoated samples. Therefore, the natural active components of EOs showed positive effects on extending the shelf-life of coated chicken fillets. Besides, sensory properties of coated samples were significantly ($p < 0.05$) better than those of control samples. Generally, TEO was more effective than SEO and no synergistic effect was observed between TEO and SEO.

1. Introduction

In recent years, the consumption of chicken and poultry has remarkably increased due to the low cost of production compared to other meat products, low amounts of fat and an important source of protein and other nutritional compounds (Noori, Zeinali, & Almasi, 2018). However, the fresh poultry is susceptible to microbial growth and lipid oxidation due to high protein and moisture content, high pH close to natural and aerobic conditions during storage which reduces the shelf-life to approximately 4–5 days (Kerry, O'grady, & Hogan, 2006; Latou, Mexis, Badeka, Kontakos, & Kontominas, 2014). In addition to microbial spoilage, fresh poultry is also prone to contamination by pathogenic bacteria such as *Salmonella* spp., *Campylobacter jejuni*, *Escherichia coli* O157: H7, *Listeria monocytogenes*, and so on

(Woraprayote et al., 2016). Preventing microbial growth and delaying lipid oxidation can increase the shelf-life of preserved poultry.

Nowadays, a major challenge in the poultry industry is increasing the shelf-life of fresh meat. In this regard, the application of chemical preservatives such as potassium sorbate, benzoic acid, and sodium nitrate have adverse effects on human health. Therefore, recent studies have been focused on the use of natural preservatives such as essential oils (Takma & Korel, 2019), organic acids (Mexis, Chouliara, & Kontominas, 2012), bacteriocins (Raeisi, Tabaraei, Hashemi, & Behnampour, 2016), chitosan and modified atmosphere packaging (Latou et al., 2014) to extend the shelf-life and retard the lipid oxidation of fresh poultry. Among natural antimicrobial compounds, plant-derived essential oils (EOs) have been widely investigated (Dehghani, Hosseini, Golmakani, Majdinasab, & Esteghlal, 2018; Lin, Gu, Sun, & Cui, 2019; Lin, Zhu, &

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Cui, 2018; Takma & Korel, 2019). EOs are aromatic and volatile oily liquids obtained from different parts of plants including buds, flowers, leaves, seeds, roots, bark and fruits through methods such as fermentation, expression, extraction or steam distillation (Burt, 2004). They are generally recognized as safe (GRAS) by the Food and Drug Administration (FDA) and commonly used as flavoring agents in the food industry. Moreover, the antimicrobial and antioxidant activities of some EOs have long been recognized. These functional characteristics are related to phenolic and terpenoid compounds presented in EOs structure (Burt, 2004). Incorporation of EOs into biopolymer-based edible films and coatings is a new approach to produce antimicrobial and antioxidant coatings and films for extending the shelf-life of foodstuffs such as meat, fish and poultry products (Bazargani-Gilani, Aliakbarlu, & Tajik, 2015; Javaherzadeh, Bafroee, & Kanjari, 2020; Karimian, Bafroee, & Sharifan, 2019; Sharma et al., 2017; Umaraw & Verma, 2017). In this method, active ingredients are released into the food surface at a controlled level. A variety of antimicrobial edible coatings incorporated with EOs such as oregano (Artiga-Artigas, Acevedo-Fani, & Martín-Belloso, 2017; Hashemi & Khaneghah, 2017), ginger (Noori et al., 2018), clove (Mulla et al., 2017), and black cumin oil (Takma & Korel, 2019) have been developed on chicken fillets. Javaherzadeh et al. (2020), investigated the films containing *Polylophium involucreatum* EO which could prolong the shelf-life of chicken fillets during storage by controlling the increase of pH, TVBN, and PV and reducing the microbial growth. Coatings containing clove, holy basil, cassia, and thyme EOs showed lower values of pH, TBARS and microbial population vacuum packaged fresh chicken sausages during storage at -18 ± 2 °C for 45 days (Sharma et al., 2017).

Among them, Basil (*Ocimum basilicum* L.) as a pharmaceutical and aromatic herb has attracted a lot of attention to be used as an edible film, especially in Iran. A notable layer of mucilage is formed around the Basil seeds when they are soaked in water (Hashemi & Khaneghah, 2017). Basil-seed gum (BSG) is comprised of two main components including an acid-stable core glucomannan and a-linked xylan with acidic side chains at C-2 and C-3 of the xylosyl residues in the acid-soluble portion. BSG offers some functional properties including biodegradability, hydrophilicity, low production cost, heat-resistant, antimicrobial, and antioxidant properties. Furthermore, excellent rheological characteristics of the extracted gum make it a good candidate to produce edible coatings and films (Hashemi & Khaneghah, 2017; Naji-Tabasi & Razavi, 2017).

Summer savory (*Satureja hortensis*) is commonly used worldwide as an herbal plant with great medical benefits. Summer savory EO (SEO) shows good antimicrobial and antioxidant properties due to the presence of high contents of volatile active compounds including carvacrol, thymol, *p*-cymene, and γ -terpinene (Abdollahi, Damirchi, Shafafi, Rezaei, & Ariaii, 2019).

Shirazi thyme (*Zataria multiflora*) is a member of the Lamiaceae family that grows in southern and central of Iran (Dehghani et al., 2018). Shirazi thyme essential oil (TEO) has great antioxidant and antimicrobial activities due to the high concentration of oxygenated phenolic monoterpenes. The main active components of the TEO are thymol and carvacrol (Raeisi, Tajik, Aliakbarlu, Mirhosseini, & Hosseini, 2015).

High antioxidant and antimicrobial properties of TEO and SEO make them good candidates for developing antimicrobial active packaging in combination with biopolymers. However, the oily nature of EOs can affect the physicochemical characteristics of hydrophilic biopolymers by changing their integrity or degree of hydrophobicity. Therefore, it is essential to find the optimum concentration of EO with minimum effect on the biopolymer properties while supplying the desired functional properties (Abdollahi et al., 2019).

With reference to the subject mentioned above, the present study aimed to develop a new active antimicrobial coating based on BSG incorporated with emulsions of TEO and SEO at different proportions in order to extend the shelf-life of chicken fillets during cold storage. EOs were incorporated individually or in combination to investigate any possible synergistic effect. To our knowledge, the application of BSG

incorporated with TEO and SEO emulsions in the preparation of antimicrobial coating for shelf life extension of chicken fillets has not been studied yet. We made an effort to find applicability of BSG as a low cost and available polysaccharide in combination with antimicrobial EOs to increase quality and shelf life of meat products.

2. Materials and methods

2.1. Chemicals

Standard Plate Count Agar (PCA), de Man Ragosa Sharpe Agar (MRSA), peptone water, sodium chloride, magnesium oxide, filter paper, methanol, paraffin, boric acid, methyl red, sulfuric acid, Tween 80, 1-butanol, thiobarbituric acid (TBA), glacial acetic acid, chloroform, glycerol, potassium iodide and sodium thiosulfate were purchased from Merk Co. (Darmstadt, Germany). Hexane and ethanol (96% v/v) were obtained from Dr. Mojallali Chemical Laboratories (Tehran, Iran). BSG was purchased from Reyhan Gum Parsian (Gorgan, Iran).

2.2. Preparation of chicken fillets

Fresh skinless chicken breast fillets were purchased from a local market in Shiraz (Fars, Iran) and immediately transferred to the laboratory in insulated iceboxes. Chicken fillets were cut manually into cubes of approximately equal weight using a sterilized knife and kept at 4 °C before coating.

2.3. Essential oil extraction

Shirazi thyme and savory plants containing essential oils were purchased from local grocery and authenticated at the School of Agriculture, Shiraz University (Shiraz, Iran). The hydro-distillation method using a Clevenger-type apparatus was employed to extract the EOs for 3 h. The distillation process was followed by centrifugation and dehydration of EOs using anhydrous Na₂SO₄. Then, EOs were kept at 4 °C in dark bottles (Dehghani et al., 2018).

2.4. Preparation of BSG coating dispersion

The amount of 11.25 g BSG was dissolved into 1200 ml distilled water while stirring at room temperature for 4 h. The hydration process was completed at 4 °C for 24 h. Coating dispersion was prepared by dissolving glycerol (30% wt of BSG) into 0.75% (v/v) BSG dispersion under constant stirring for 30 min. TEO and SEO were then added individually or in combination with a ratio of 1:1 to reach a final concentration of 1 and 2 (% v/v). Before adding EOs, Tween 80 was added as an emulsifier with the same concentration of EO and mixed for 30 min. The dispersion was then homogenized using a homogenizer (HG-15D, Daihan Scientific, Korea) at 15,000 rpm for 5 min. Final coating dispersions were obtained after sonication for 5 min using a SONOPULS ultrasonic homogenizer (HD3200, 150 W, 20 KHz, Bandelin electronic GmbH & Co., Berlin, Germany). BSG dispersion without the added EO was obtained by the same method and used as control samples.

2.5. Chicken fillets coating method

The coating was performed by dipping the fillets into the coating dispersion for 120 s. Samples were placed on a grind for 2 min at room temperature in order to drain. The coating procedure was repeated after 2 min and the excess dispersion was drained off. The coating layer was developed at room temperature during 1 h. All samples were placed in clear polystyrene takeout containers and kept at 4 ± 1 °C for 12 days. Sampling was performed on days 1, 4, 8, 10, and 12 of storage. In this study, the effects of eight treatments were investigated: control 1 (non-coated chicken filets); control 2 (coated fillets without EOs in coating dispersion); TEO 1% (coated fillets containing 1% TEO in coating

dispersion); TEO 2% (coated fillets containing 2% TEO in coating dispersion); SEO 1% (coated fillets containing 1% SEO in coating dispersion); SEO 2% (coated fillets containing 2% SEO in coating dispersion); TEO + SEO 1% (coated fillets containing 1% combination of TEO and SEO (1:1 ratio) in coating dispersion); and TEO + SEO 2% (coated fillets containing 2% combination of TEO and SEO (1:1 ratio) in coating dispersion).

2.6. Chemical analyses

2.6.1. Measurement of chicken fillets composition

Proximate composition of chicken fillets including moisture and crude ash contents was determined at 105 °C and 550 °C, respectively (AOAC, 2005). Total crude protein was measured using the Kjeldahl method (AOAC, 2005). The lipid content was determined according to the method of Bligh and Dyer (1959).

2.6.2. Measurement of EOs compounds through GC/MS analysis

Analysis and quantification of EOs' components were performed using a Beifen 3420A gas chromatograph (Beijing Beifen-Ruili Analytical Instrument Co., Ltd., Beijing, China) equipped with a split/splitless injector, a flame ionization detector and a BPX5 capillary column (5% Phenyl Polysilphenylene-siloxane, 30 m × 0.25 mm internal diameter; 0.25 µm film thickness, SGE Analytical Science, Melbourne, Australia). The initial temperature of column was 60 °C, followed by raising to 210 °C at a rate of 3 °C/min; afterward, it was raised to 240 °C at 20 °C/min and maintained for 8.5 min. Identification of EOs components was performed through comparison to the obtained retention profile with those of known samples already analyzed by a gas chromatograph (Agilent 7890A) coupled to a mass spectrometer (Agilent 5975C) with the similar column and operating conditions.

2.6.3. Measurement of EOs antioxidant activity

The antioxidant activity of extracted EOs was evaluated by measuring the scavenging activity of 2,2-diphenyl-1-picrylhydrazyl (DPPH) free radicals according to the method of Albayrak et al. (Albayrak, Aksoy, Sagdic, & Hamzaoglu, 2010). Different concentrations of EOs in methanol were prepared. The methanolic solution of DPPH (0.1 mM) was mixed with the methanolic solution of EOs at different mixing ratios. EO/DPPH mixtures were incubated in darkness for 1 h to complete the reaction. After that, the absorbance of the samples was measured at 517 nm against methanol as the blank. The control sample was prepared by mixing DPPH solution and methanol. The amount of EO leading to 50% inhibition (IC₅₀) was determined from absorbance versus concentration profiles. The percentage in inhibition of the DPPH radical was calculated using the following equation (1).

$$\text{DPPH scavenging capacity (\%)} = [(A_c - A_s)/A_c] \times 100 \quad (1)$$

Where A_c is the absorbance of the control sample and A_s is the absorbance of the treated sample.

2.6.4. pH measurement

2 g of fillet sample was homogenized in 90 ml of distilled water and stirred for 30 s. After filtration, the pH value of filtrate was recorded using a digital pH meter (STARTER 3000, OHAUS Co., Switzerland).

2.6.5. Determination of total volatile basic nitrogen (TVB-N)

TVB-N index was measured according to the method described by Ojagh et al. (Ojagh, Rezaei, Razavi, & Hosseini, 2010). Briefly, a mixture of homogenized samples and magnesium oxide was subjected to distillation by a Kjeldahl type apparatus (Kjeltec PDU-500, PECO Co., Iran). The distillate was collected in an aqueous solution of 3% boric acid containing a mixed indicator of 0.1 g methyl red and 0.1 g methylene blue (dissolved in 100 ml ethanol). The boric acid solution was then titrated with 0.05 M sulfuric acid solution. TVB-N values were

determined in mg N₂ per 100 g of chicken fillet.

2.6.6. Determination of peroxide value (PV)

PV was measured in the total lipid extracts using the method of Pearson (Egan, Kirk, & Sawyer, 1997, pp. 609–634). Briefly, after the lipid extraction of the samples, 3 g of lipid was mixed with 30 ml of glacial acetic acid/chloroform mixture solution (mixing ratio 3:2, respectively) and 0.5 ml of saturated potassium iodide. The mixture was kept in dark conditions for 10 min. After that, 30 ml of distilled water and 0.5 ml of freshly prepared starch suspension (1%) were added to the mixture. The mixture was shaken and titrated with 0.01 M sodium thiosulfate until the color disappeared. Results were expressed in milliequivalents of active oxygen/kg of lipid according to the following equation (2):

$$\text{PV} = \frac{N \times V}{W} \times 100 \quad (2)$$

where N is the normality of sodium thiosulfate, V is the volume of sodium thiosulfate, and W is the weight of fat.

2.6.7. Determination of thiobarbituric acid reactive substance (TBARS)

The thiobarbituric acid values of samples were measured colorimetrically by the method of Kirk and Sawyer (Kirk & Sawyer, 1991, pp. 640–643). Briefly, 200 mg of homogenized sample was mixed into 25 ml of 1-butanol. Five ml of the mixture was pipetted into a dry stoppered test tube and mixed with 5.0 ml of thiobarbituric reactive reagent (prepared by dissolving 200 mg 2-TBA in 100 ml of 1-butanol, filtered and stored at 4 °C for not more than 5 days). The test tube was placed in a boiling water bath for 120 min and then cooled in an ice bath for 10 min. The absorbance of the sample (A_s) was measured at 532 nm against water as the blank. A control sample was also prepared and treated in the same procedure. TBARS values in mg of malondialdehyde equivalents per kg of tissue were calculated by the following equation (3).

$$\text{TBARS} = 50 \times [(A_s - A_b) / 20] \quad (3)$$

Where A_b is the absorbance of the control sample and A_s is the absorbance of the treated sample.

2.7. Microbiological analysis

Ninety milliliters of sterile saline was added to 10 g of fillet samples and homogenized by a stomacher to prepare serial dilutions for pour plate method cultivation. The sample dilutions poured on plate count agar (PAC) media were incubated at 37 °C for 48 h to determine the total viable mesophilic bacteria count. Psychrotrophic bacteria counts were determined on PCA media after 10 days of incubation at 7 °C, while lactic acid bacteria (LAB) growth was measured on MRS agar media incubated at 25 °C under anaerobic conditions for 5 days (Dehghani et al., 2018; Raeisi et al., 2015). The results were reported in logarithms of colony-forming units (CFU) per gram of sample (log₁₀ CFU/g).

2.8. Cooking loss

To estimate the cooking loss, 30 g of fillet sample was cooked by steam in a water bath at 80 °C for 1 h. After cooking, the sample was weighed again. The percentage of weight loss relative to the initial weight was determined (Barbanti & Pasquini, 2005).

2.9. Sensory evaluation

Organoleptic characteristics of chicken fillets were evaluated by a 12-member qualified panel according to the method described by Hernández et al. (Hernández et al., 2009) and Noori et al., (Noori et al., 2018). All panelists were familiar with main sensory evaluation techniques and spoilage indicators of fresh chicken fillets. A 5-point hedonic

scale (1, dislike extremely to 5, like extremely) was utilized to score the samples for texture, odor, color, and overall acceptability. Chicken fillets with overall scores of more than 4 were considered to be acceptable.

2.10. Statistical analysis

All experiments were performed at least three times and analyzed using factorial test by SAS (ver. 9.1, SAS Institute Inc., Cary, NC) at a confidence level of 0.05. First, the Kolmogorov-Smirnov test was used for data normalization and then homogeneity of data variance was investigated by the Lyon test. The results of these tests were used to analyze the data of the experimental treatments. Tukey's test was used to compare the results of physicochemical and microbiological analysis and a significant difference was considered at $p < 0.05$. The 2015 Excel software was used to draw charts.

3. Results and discussion

3.1. Identification of chicken components

The results of the chicken fillet analysis including moisture, protein, fat, and ash content (%) were 70.1 ± 0.4 , 21.4 ± 0.12 , 5.1 ± 0.1 and 1.2 ± 0.08 , respectively. These results were consistent with the results of Noori et al. (Noori et al., 2018), and Bazargani-Gilani et al. (Bazargani-Gilani et al., 2015). The differences between the proximate compositions of chicken meat in different studies can be greatly correlated with the chicken diet, chicken size, chicken age, season, breeding conditions, and other environmental conditions. These differences can lead to changes in microbial growth and chemical corruption, as well as sensory properties such as texture, odor, color, and overall acceptability (Nowzari, Shábanpour, & Ojagh, 2013).

3.2. Identification of essential oils components

The chemical components of TEO and SEO were measured by GC-MS (Table 1). Twenty-one compounds were identified in TEO. All of the identified compounds exhibited 98.78% of total TEO. The major representative components were thymol (41.02%) and carvacrol (29.58%). The *p*-cymene (7.41%), α -terpinene (6.02%) and α -pinene (2.21%) were the other major constituents. Among these compounds, carvacrol and thymol possess high antioxidant properties and also

carvacrol, thymol, α -pinene, and paracetamol exhibit antimicrobial properties. The savory plant contains various compounds. After analysis of the SEO, 18 compounds with different percentages were identified. All of these compounds represented 96.90% of total SEO. Carvacrol (39.28%) and γ -terpinene (40.93%) were the major constituents. These compounds are known to have strong antioxidant effects (Hamidpour, Hamidpour, Hamidpour, Shahlari, & Sohraby, 2014). The differences in main chemical compounds of TEO and SEO and their amounts determined in other references can be related to age and stage of maturity, genetic, soil composition, weather conditions, plant organs, distillation conditions and other factors (Rehman, Hanif, Mushtaq, & Al-Sadi, 2016).

3.3. The antioxidant properties of EOs

Phenolic compounds (or polyphenols), present in essential oils, have been known to possess antioxidant activities. The important role of phenolic compounds is to scavenge free radicals (Sharififar, Moshafi, Mansouri, Khodashenas, & Khoshnoodi, 2007). Through the DPPH scavenging assay, the amount of thyme essential oil leading to 50% inhibition (IC_{50}) was 208.83 ppm, while the IC_{50} of savory essential oil was 340.50 ppm. This indicated that less concentration of TEO leads to 50% inhibition of DPPH free radicals. Therefore, thyme EO shows greater antioxidant activity due to high content of effective components such as thymol, carvacrol and *p*-cymene. Antioxidant properties of thymol and carvacrol were reported previously (Ruberto & Baratta, 2000). The synergistic effect between main components and other rare compounds found in TEO can significantly increase the scavenging capacity of TEO compared to SEO.

3.4. Effect of BSG-based antimicrobial coatings on the pH value of chicken fillets

In this study, the initial pH of chicken fillet was 5.88, which was in agreement with previous works (Abdou, Galhoum, & Mohamed, 2018). As shown in Fig. 1, the pH values of samples increased during the storage period. The final pH of the fillet samples was obtained through the balance of alkaline compounds and the lactic acid produced by bacteria during storage period. Non-significant ($p > 0.05$) changes were observed in pH until the 4th day of storage; however, it was increased after that. The pH of uncoated samples decreased in the 8th day of storage,

Table 1
Chemical composition of TEO and SEO measured by GC-MS.

Number	TEO			SEO		
	Compound	Retention time (min)	Concentration (Peak area %)	Compound	Retention time (min)	Concentration (Peak area %)
1	α -thujene	4.31	0.20	α -thujene	1.89	1.91
2	α -pinene	4.52	2.21	α -pinene	1.88	1.50
3	Camphene	5.47	0.16	Camphene	1.87	1.10
4	β -pinene	6.74	0.85	Sabinene	1.75	0.09
5	3-octanone	5.75	0.35	Myrcene	3.48	2.47
6	Myrcene	7.85	1.12	β -pinene	2.16	0.79
7	α -terpinene	8.91	6.02	α -phellandrene	3.61	0.28
8	<i>p</i> -cymene	7.86	7.41	α -terpinene	5.40	4.03
9	Limonene	8.84	1.67	Limonene	19.48	0.34
10	γ -terpinene	8.82	0.34	β -phellandrene	21.16	0.32
11	Terpinolene	14.14	0.55	<i>p</i> -cymene	26.51	2.24
12	Linalool	15.03	1.12	γ -terpinene	33.51	40.93
13	Thymol methyl ether	14.34	0.14	Terpinene-4-ol	8.92	0.12
14	Carvacrol methyl ether	15.77	1.63	α -terpineol	7.57	0.05
15	Carvacrol	19.38	29.58	Carvacrol	32.01	39.28
16	Thymol	17.98	41.02	β -caryophyllene	12.01	0.82
17	Thymol acetate	23.09	2.46	β -bisabolene	10.01	0.62
18	Carvacrol acetate	26.13	0.29	Spathulenol	1.12	0.01
19	(E)-caryophyllene	30.17	0.49			
20	Viridiflorene	31.74	0.62			
21	Aromadendrene	32.27	0.55			
Total			98.78			96.90

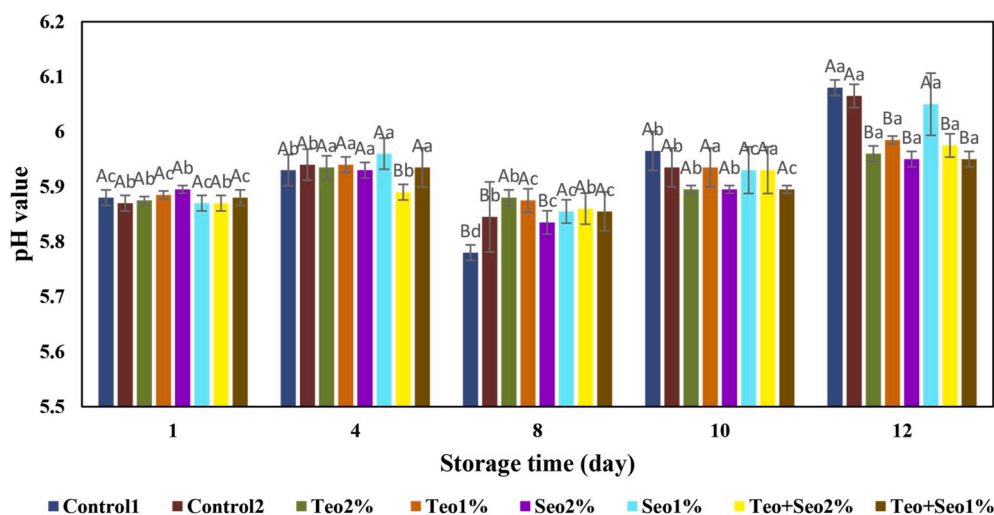


Fig. 1. Changes in pH values of chicken fillet samples during storage at 1 ± 4 °C; Control 1: uncoated samples; Control 2: EO-free BSG-coated samples; Teo: coated samples with BSG and Shirazi thyme essential oil emulsion; Seo: coated samples with BSG and savory essential oil emulsion. Significant difference ($p < 0.05$) between data is expressed by different letters; capital letters show significant difference of various treatments through the same storage period, and lower-cased letters indicate significant difference among the same treatments during the storage period.

probably due to the presence of lactic acid produced by lactic acid bacteria. After 8 days, the pH of all samples increased in different rates. The increase in pH was due to the formation of basic nitrogenous compounds such as ammonia and biogenic amines as a result of the autolytic activity of indigenous enzymes and the proteolytic activity of psychrotrophic bacteria which can provide the context for chemical spoilage (Badee, Moawad, ElNoketi, & Gouda, 2013). At the end of storage, the highest pH was measured in uncoated samples (6.08) and the lowest pH was reached in coated fillets containing 2% SEO (5.95) and 2% TEO (pH 5.96). There were no significant differences ($p > 0.05$) in the pH values among BSG coated samples containing 2% TEO, 2% SEO, 2% TEO + SEO and 1% TEO on day 12. However, an increase in the EO concentration represented more inhibitory effects through reducing the microbial growth during storage period. Samples coated with BSG and EOs showed lower pH and longer shelf-life due to the effect of coating dispersions on microbial activity. Compared to TEO, SEO had less impact on the pH values of chicken samples during storage, indicating less antimicrobial activity of SEO than TEO. In this study, no

significant synergistic effect was observed between TEO and SEO. The strong antimicrobial activity of TEO has been previously reported (Dehghani et al., 2018).

3.5. Effect of BSG-based antimicrobial coatings on the TVB-N index of chicken fillets

TVB-N index can be considered one of the main chemical indicators for shelf-life of meat and meat products, which mainly includes ammonia and primary, secondary, and tertiary amines (Ala & Shahbazi, 2019). Changes in the TVB-N values of chicken fillets during storage are presented in Fig. 2. In this study, the average initial TVB-N value of the chicken fillet samples in the 1st day was 8.24 mg N/100 g. The TVB-N of all treatments was progressively increased during cold storage. After 12 days, TVB-N content of uncoated control sample reached 29.42 mg N/100 g. According to the acceptability limit of TVB-N (25 mg N/100 g) for initiation of microbial spoilage and deterioration of meat and meat products, uncoated control samples and control samples with only BSG

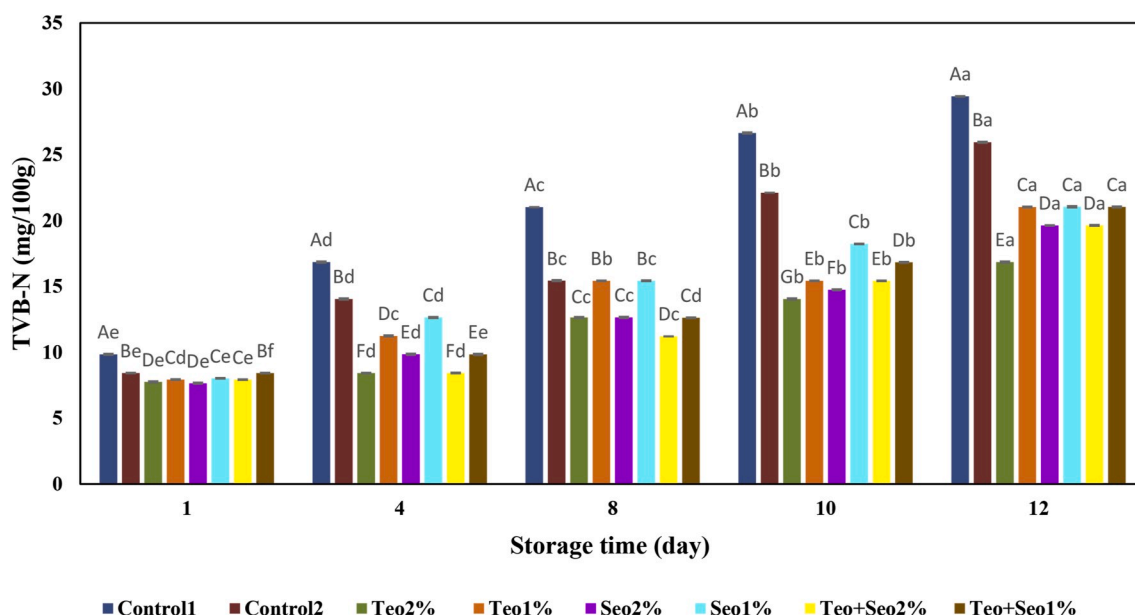


Fig. 2. Changes in TVB-N values of chicken fillet samples during storage at 1 ± 4 °C; Control 1: uncoated samples; Control 2: EO-free BSG-coated samples; Teo: coated samples with BSG and Shirazi thyme essential oil emulsion; Seo: coated samples with BSG and savory essential oil emulsion. Significant difference ($p < 0.05$) between data is expressed by different letters; capital letters show significant difference of various treatments through the same storage period, and lower-cased letters indicate significant difference among the same treatments during the storage period.

coating exceeded this limit on days 10 and 12 respectively. According to the results, the BSG coating without essential oil cannot be effective to reduce the production of volatile nitrogen bases. A significant ($p < 0.05$) decrease in TVB-N contents was observed in samples coated with EOs-incorporated BSG. The TVB-N values of 16.84, 21.03, 19.62, 21.04, 19.63 and 21.04 mg/100g were recorded for the 2% TEO, 1% TEO, 2% SEO, 1% SEO, 2% TEO + SEO and 1% TEO + SEO treatments, respectively, versus 29.42 and 25.94 mg/100g for the control samples on day 12. It can be observed that TVB-N content of coated samples containing EOs in coated dispersion was not beyond the acceptable limit after 12 days of storage. Moreover, by increasing the concentration of essential oils, TVB-N values of treated samples raised more slowly. At the end of storage, the lowest amount of TVB-N was measured in coated fillets containing 2% TEO. In general, at any time of storage BSG-EOs coated samples showed lower TVB-N content compared to control samples. However, a significant synergistic effect was not found between TEO and SEO. The decrease in the bacterial count or their capacity for decomposition of chicken protein in samples containing polyphenols can be a probable explanation of the lower TVB-N values of coated fillets with EOs in comparison with control ones (Ojagh et al., 2010). Considering the acceptable limit of TVB-N index, the shelf-life of controls and EOs-BSG coated samples were 10 and more than 12 days, respectively. Similar results were reported by Abdou et al. (Abdou et al., 2018).

3.6. Effect of BSG-based antimicrobial coatings on the peroxide value (PV) of chicken fillets

Hydroperoxides are primary products of the lipid oxidation process. Decomposition of chemically active hydroperoxides leads to the production of secondary oxidative products, and undesirable odor and taste of meat products. The PV indicates the oxidative progress by measuring these primary products. Fig. 3 shows the PVs of control and treated chicken fillets during the storage at 4 °C for 12 days. Generally, the PVs increased significantly during storage for all samples. The initial PV of uncoated control samples was 1.21 meq active O₂/kg lipid and reached to maximum amount of 8.07 meq active O₂/kg lipid on day 12 of storage. In uncoated control samples and free-EO BSG coated samples, formation of hydroperoxides was accelerated after 8 days and 12 days, respectively, and exceeded the acceptable limit (5 meq active O₂/kg lipid). Hydroperoxide formation in BSG coated control samples was lower than in uncoated control fillet samples. This can be related to the oxygen barrier characteristics of BSG-based coatings. Formation of hydroperoxides further decreased in EOs-incorporated BSG coated samples due to antioxidant properties of TEO and SEO. The lowest amount of PV

was found in coated fillets containing 2% TEO in coating dispersion. This can be due to the higher amounts of phenolic compounds such as thymol and carvacrol in thyme compared to savory and also the synergistic effect of these compounds with other trace components of TEO. Generally, EOs are a source of phenolic and polyphenolic components with strong antioxidant characteristics (Abdou et al., 2018). The antioxidant activity of EOs is related to the several mechanisms including interaction with free radicals (due to strong radical scavenging properties of phenolic compounds), inhibition of chain reactions, decomposition of hydroperoxides, and binding of transition metals. Actually, resonant structures of phenolic radicals lead to their thermodynamic stability (Dehghani et al., 2018).

3.7. Effect of BSG-based antimicrobial coatings on the TBARS values of chicken fillets

Generally, hydroperoxides are very unstable compounds and do not contribute to the improper sensory characteristics prior to the development of secondary products of lipid oxidation. Therefore, TBARS value represents the secondary products of lipid oxidation particularly aldehydes (Cai et al., 2014). In addition to malondialdehyde, other aldehydes are able to produce red pigments after reaction with TBA. Therefore, the amount of secondary oxidation products is expressed in TBARS value (mg malonaldehyde equivalents per kg of sample or mg MDA eq/kg). The maximum tolerable limit of TBARS without any adverse effect on chicken quality is 4 mg MDA eq/kg. As shown in Fig. 4, the TBARS values of chicken fillets ranged from 0.25 to 0.19 mg MDA/kg in the 1st day. During storage, TBARS values of all samples gradually increased. The TBARS values of all samples were below the maximum limit until the 10th day. At the end of storage (day 12), TBARS values of control samples exceeded the acceptable limit. However, BSG coated samples (control 2) had a lower TBARS compared to uncoated samples (control 1). Also, BSG-based coated samples containing EOs showed significantly ($p < 0.05$) lower amounts of TBARS in the acceptable limit compared to the control samples during the whole period storing time. At the end of storage period, the highest and lowest amount of TBARS values were recorded for control 1 samples (5.31 mg MDA/kg) and coated fillets containing 2% TEO (2.44 mg MDA/kg), respectively. As it mentioned earlier, the antioxidant activity of SEO was weaker than those of TEO.

The results emphasized that coating of chicken fillets with BSG and BSG incorporated with EOs leads to oxidation retardation and increases the shelf-life of chicken fillets. This can be attributed to the protecting effects of coating against oxygen and the antioxidant activity of phenolic

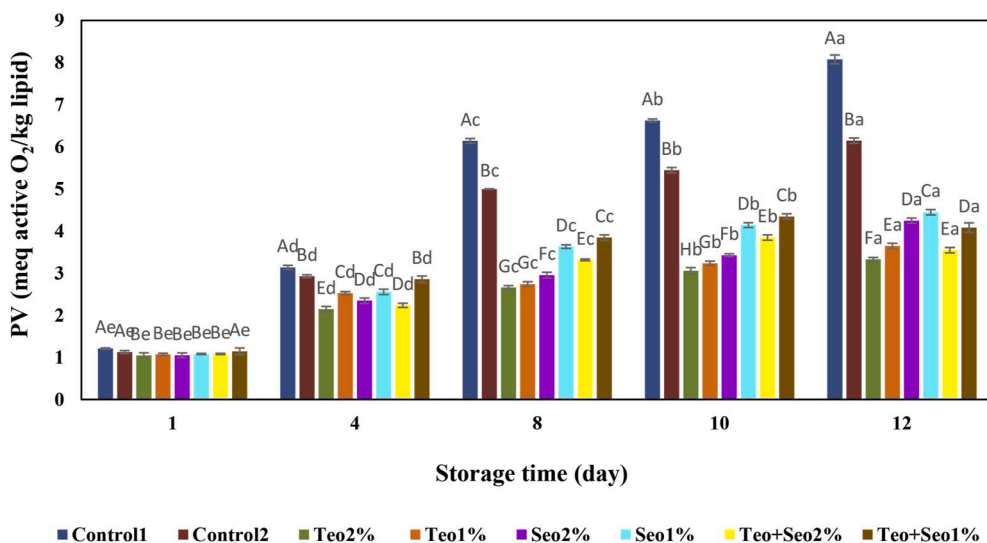


Fig. 3. Changes in PV values of chicken fillet samples during storage at 1 ± 4 °C; Control 1: uncoated samples; Control 2: EO-free BSG-coated samples; Teo: coated samples with BSG and Shirazi thyme essential oil emulsion; Seo: coated samples with BSG and savory essential oil emulsion. Significant difference ($p < 0.05$) between data is expressed by different letters; capital letters show significant difference of various treatments through the same storage period, and lower-cased letters indicate significant difference among the same treatments during the storage period.

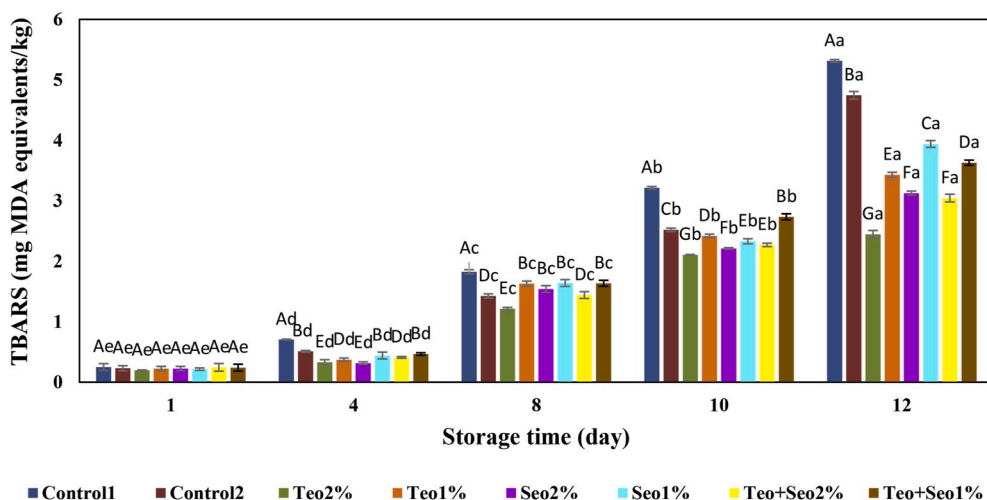


Fig. 4. Changes in TBARS values of chicken fillet samples during storage at 1 ± 4 °C; Control 1: uncoated samples; Control 2: EO-free BSG-coated samples; Teo: coated samples with BSG and Shirazi thyme essential oil emulsion; Seo: coated samples with BSG and savory essential oil emulsion. Significant difference ($p < 0.05$) between data is expressed by different letters; capital letters show significant difference of various treatments through the same storage period, and lower-cased letters indicate significant difference among the same treatments during the storage period.

compounds presented in TEO and SEO. Not only coatings are effective barriers against oxygen, but also the EO emulsions significantly ($p < 0.05$) enhanced the antioxidant properties of BSG-based coatings. Therefore, the development of oxidation products in EO-free coated samples was lower than non-coated ones, which is attributed to the protecting effect of BSG coating.

3.8. Effect of BSG-based antimicrobial coatings on the microbiological quality of chicken fillets

3.8.1. Mesophilic bacteria

Fresh or minimally processed meat products are very susceptible to microbial contamination during the various steps of production to consumption. Therefore, determining the mesophilic bacteria growth is necessary in most safety standards (Dehghani et al., 2018). Changes in the total viable count (TVC) of chicken fillet samples as a function of storage time at 4 °C are shown in Fig. 5a. The initial TVC of chicken fillets was ≈ 4.5 log CFU/g which was in agreement with the previous results for fresh chicken meat (Bazargani-Gilani et al., 2015; Economou, Pournis, Ntzimani, & Savvaidis, 2009). The maximum acceptable limit of TVC for fresh meat is reported 7 log CFU/g (Senter, Arnold, & Chew, 2000). Although TVCs of samples increased during storage in a treatment-dependent manner, it remained below the acceptable limit until the 8th day of storage. At day 12, TVC of uncoated samples and EOs-free coated samples exceeded to 7.96 and 7.49 log CFU/g, respectively, which were significantly ($p < 0.05$) higher than the coated samples with BSG containing TEO and SEO.

Whereas the BSG played the role of an oxygen barrier, the antimicrobial effects of coating enhanced by increasing the concentration of essential oils due to the presence of phenolic compounds, terpenoids, saponin, aldehydes and tannins. These compounds inhibited the microbial growth in samples treated with EOs by affecting the cell membrane. In this regard, thymol, carvacrol, and γ -terpinene are the major monoterpene phenols in TEO and SEO which have antimicrobial activity against a wide spectrum of microorganisms (Jouki, Yazdi, Mor-tazavi, Koocheki, & Khazaei, 2014). They are able to disintegrate the outer membrane of Gram-negative bacteria, leading to release lipopolysaccharides and to increase the permeability of the cytoplasmic membrane to ATP energy source. Furthermore, these compounds work as an antimicrobial agent against Gram-positive bacteria likely through increasing the membrane permeability and fluidity, and the inclusion between fatty acid chains, which leads to the formation of channels through the membrane and increases permeability (Burt, 2004).

The results of this study showed no significant difference in TVCs between uncoated and EOs-free coated samples up to the 8th day of storage, which indicates the ineffectiveness of EOs-free BSG coatings

against mesophilic bacteria ($p < 0.05$). However, in the 10th and 12th days of storage TVC increased significantly in uncoated samples. Based on the results of TVC, it could be clearly found that TEO/BSG coatings exhibited the highest antimicrobial activity against total bacterial count. In support of previous studies, BSG-based coatings containing TEO leads to a remarkable inhibition of microorganisms' growth in chicken fillet samples (~ 3.69 logs CFU/g) during 12 days at 4 °C (Bazargani-Gilani et al., 2015).

3.8.2. Psychrotrophic bacteria

Gram negative psychrotrophic bacteria such as *Pseudomonas* spp. Have been identified as the main cause of fresh meat spoilage during storage at low temperatures. Proteolytic activities of psychrotrophic bacteria are responsible for the poultry spoilage and the subsequent slime production (Barnes & Impey, 1968). The effect of edible coating and storage time on psychrotrophic bacteria count (PBC) is shown in Fig. 5b. According to the data, the initial PBC of uncoated samples was 3.62 log CFU/g and it exceeded to more than 7 log CFU/g (7.24 log CFU/g) during 12 days of storage ($p < 0.05$). While PBC of all samples followed an increasing trend during storage, the lowest amount of PBC was measured in BSG coated fillets containing 2% TEO, after 12 days. The result confirms the strong antimicrobial properties of TEO due to the high amount of thymol and carvacrol. No synergistic effect was observed between TEO and SEO in coating dispersions prepared with a mixture of EOs.

3.8.3. LAB bacteria

LABs are gram-positive bacteria that play a significant role in the chicken meat spoilage. They grow slowly at low temperatures. In this study, the effect of BSG coatings and EOs on LAB count (LABC) was investigated in chicken fillet samples stored at 4 °C for 12 days (Fig. 5c). The initial LABC in uncoated samples was 4.5 and increased to 7.6 log CFU/g after 12 days. At the end of the storage period, the lowest number of LABs was observed in coated fillets containing 2% TEO (3.8 log CFU/g). Through the results, it can be concluded that the initial reduction of microbial count in samples can be attributed to the strong antimicrobial effect of EOs' components. All these antimicrobial results may be due to the protecting effect of BSG coatings along with active compounds of TEO and SEO, and more sensitivity of Gram-positive bacteria to the phenolic compounds in comparison with Gram-negative ones. Accordingly, the direct contact of EO components with the lipid bilayer membrane of gram-positive bacteria can result in ion permeability and thus, the leakage of vital intracellular constituents (Burt, 2004).

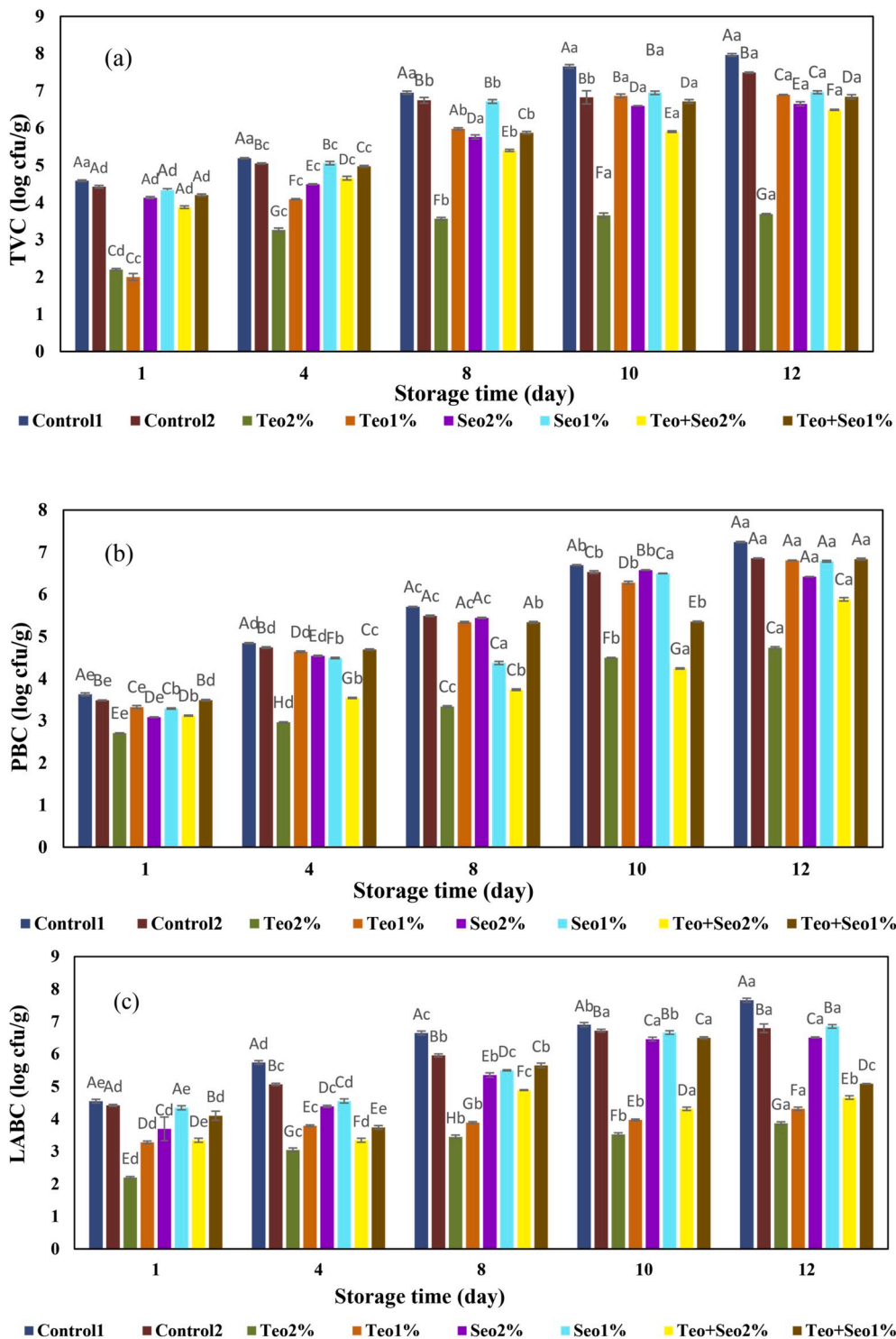


Fig. 5. Changes in (a) total viable bacterial count (TVC), (b) psychrotrophic bacterial count (PBC) and (c) lactic acid bacterial count (LABC) of chicken fillet samples during storage at $1 \pm 4^\circ\text{C}$; Control 1: uncoated samples; Control 2: EO-free BSG-coated samples; Teo: coated samples with BSG and Shirazi thyme essential oil emulsion; Seo: coated samples with BSG and savory essential oil emulsion. Significant difference ($p < 0.05$) between data is expressed by different letters; capital letters show significant difference of various treatments through the same storage period, and lower-cased letters indicate significant difference among the same treatments during the storage period.

3.9. Effect of BSG-based antimicrobial coatings on the cooking loss percentage of chicken fillets

Cooking loss is a major factor affecting the appearance and acceptance of chicken meat. Fig. 6 demonstrates the percentage of cooking loss in chicken fillet samples during 12 days of storage at 4°C . The results indicate that the cooking loss of all samples increased significantly during storage ($p < 0.05$). As reported in Fig. 6, coating the fillets with EO-free BSG could decrease the cooking loss in comparison with the uncoated samples, and cooking loss of BSG coated samples containing

EOs was significantly lower than the uncoated and EOs-free coated samples. While the cooking loss was increased about 20% in control samples from the 1st to the 12th day, this amount was only 8% in samples coated with BSG containing 2% TEO. Moreover, as it is deduced from Fig. 6, the cooking loss of the BSG/EOs coated samples is significantly ($p < 0.05$) lower than that of control and EO-free coated samples in each experiment time. It should be considered that the main part of weight loss is due to the moisture extraction. Due to the higher concentration of water on the matrix surface of control and free-EO coated samples, the water exuded easily during cooking results in greater

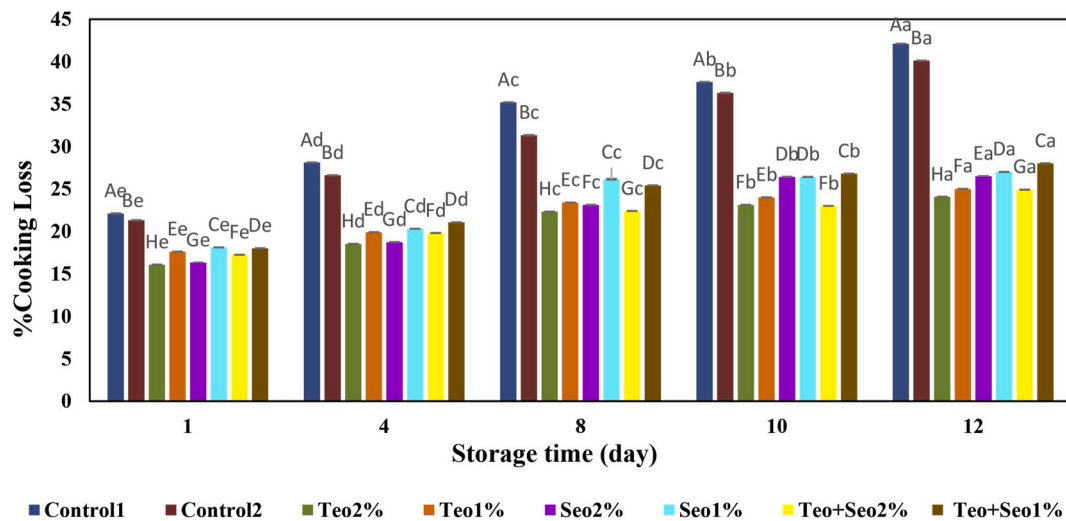


Fig. 6. Changes in percentage of cooking loss of chicken fillet samples during storage at 1 ± 4 °C; Control 1: uncoated samples; Control 2: EO-free BSG-coated samples; Teo: coated samples with BSG and Shirazi thyme essential oil emulsion; Seo: coated samples with BSG and savory essential oil emulsion. Significant difference ($p < 0.05$) between data is expressed by different letters; capital letters show significant difference of various treatments through the same storage period, and lower-cased letters indicate significant difference among the same treatments during the storage period.

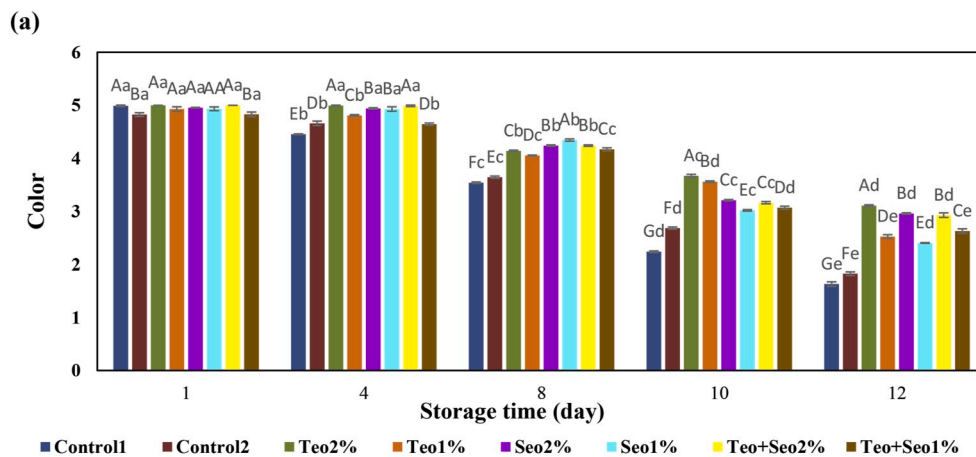
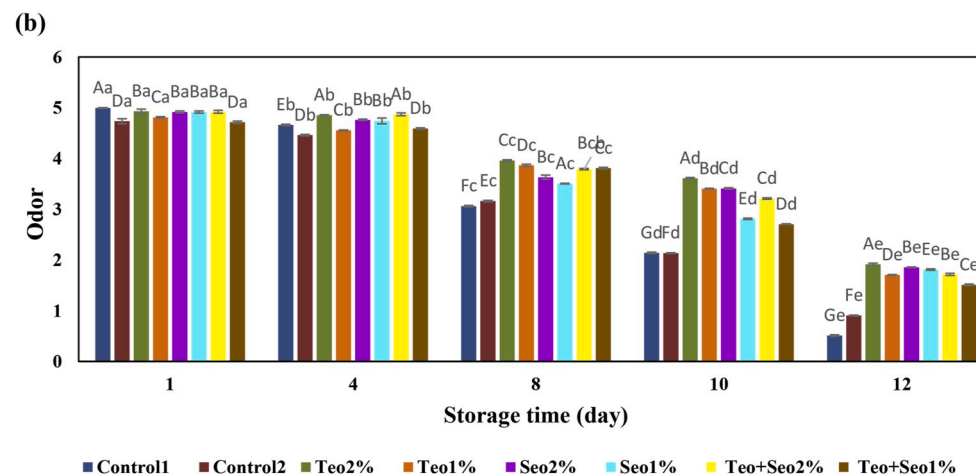
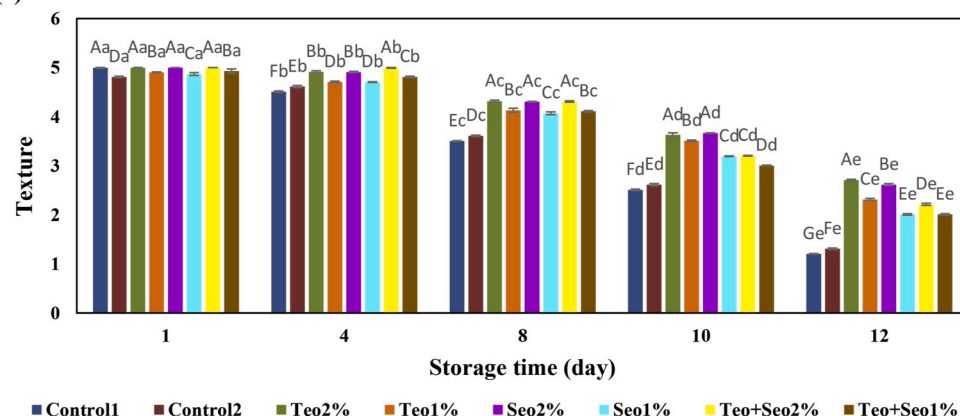


Fig. 7. Sensory evaluation of chicken fillets. Changes in color (a), odor (b), texture (c) and overall acceptability of chicken fillet samples during storage at 1 ± 4 °C; Control 1: uncoated samples; Control 2: EO-free BSG-coated samples; Teo: coated samples with BSG and Shirazi thyme essential oil emulsion; Seo: coated samples with BSG and savory essential oil emulsion. Significant difference ($p < 0.05$) between data is expressed by different letters; capital letters show significant difference of various treatments through the same storage period, and lower-cased letters indicate significant difference among the same treatments during the storage period. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

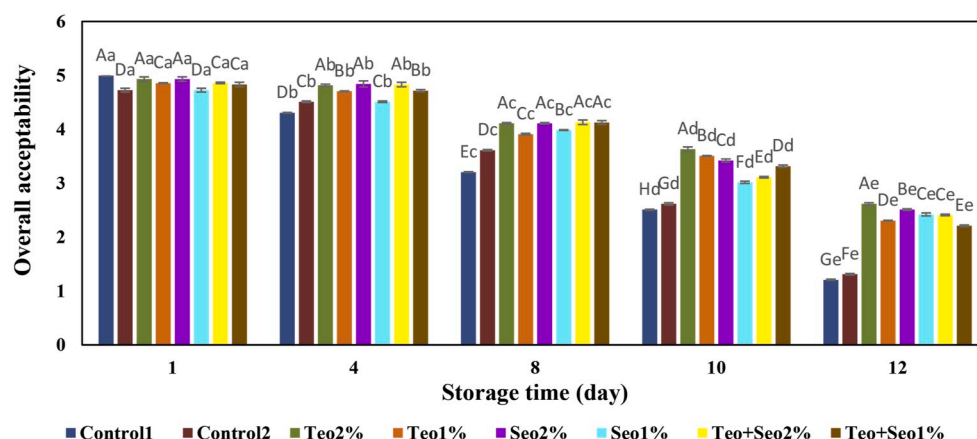


(c)

Fig. 7. (continued).



(d)



weight loss (Noori et al., 2018). However, coatings containing TEO or SEO act as a barrier to mass transfer because of the partial hydrophobicity of the surface of fillets.

3.10. Effect of BSG-based antimicrobial coatings on the sensory properties of chicken fillets

Fig. 7a, b, c and d show the sensory analysis scores of control and treated chicken fillets preserved at 4 °C for 12 days. Color, odor, texture and overall acceptability were considered in scoring by trained panelists. The chicken samples were considered to be acceptable for human consumption with a sensory score of greater than 4 (Jouki, Yazdi, Mortazavi, & Koocheki, 2014). Texture, odor, color and overall acceptability of uncoated and free-EOs coated samples obtained unacceptable score of lower than 4 by the 8th day, whereas BSG/EOs coated samples obtained unacceptable scores after 12 days. At the end of storage, the highest score was given to coated samples containing 2% TEO ($p < 0.05$). An increase in the EO concentration increased the sensory scores during evaluation. The results of the sensory evaluation were in agreement with those of microbial and chemical analyses. Due to high lipid oxidation and microbial growth, the uncoated samples showed spoilage signs such as appearing off-odor, slimy texture and discoloration after 8 days of storage. The antioxidant and antimicrobial properties of BSG/EOs coatings have been shown to minimize the oxidative effects, prolong the product shelf-life, and increase quality.

4. Conclusion

The results of this study showed that BSG-based coatings incorporated with TEO and SEO can extend the shelf-life of fresh chicken fillets through retarding the adverse chemical reactions and microbial growth. These characteristics are due to the presence of natural antimicrobial and antioxidant compounds in EOs. The use of BSG-based coatings incorporated with TEO and SEO emulsions reduced chemical and microbial deteriorations during storage (12 days in 4 °C). Various factors such as pH, TVB-N, TBARS and PV of coated samples containing EOs remained within acceptable range for a longer period.

Both TEO and SEO revealed antimicrobial properties against mesophilic, psychrotrophic and lactic acid bacteria. In this regard, Gram-negative bacteria were more resistant than Gram-positive bacteria due to the presence of outer membrane in Gram-negative bacteria. TEO showed stronger antimicrobial and antioxidant activity than SEO. No synergistic effect was found between TEO and SEO. Through this study, BSG coating has barrier properties against oxygen and provides antioxidant properties during storage. Increasing the concentration of both essential oils in coating dispersion enhanced the quality of chicken fillets during storage. The best treatment was found to be 2% TEO in coating dispersion. From an organoleptic point of view, EOs/BSG coated samples exhibited desirable sensory attributes including taste, color, odor, texture and overall acceptability by reducing lipid oxidation and microbial growth. Moreover, the EOs used in this study were extracted

from common seasonings added to foods in most cultures and induced desirable and acceptable flavor by the consumer. According to the results obtained in this study, the maximum shelf-life of 8 and 12 days could be suggested for uncoated and coated chicken fillets incorporated with EO emulsions, respectively.

Declaration of competing interest

The authors declare that there is no conflict of interest.

CRediT authorship contribution statement

Marjan Majdinasab: Supervision, Conceptualization, Methodology, Visualization, Writing - original draft. **Mehrdad Niakousari:** Supervision, Methodology. **Samaneh Shaghaghian:** Writing - review & editing. **Hamed Dehghani:** Investigation, Data curation, Software.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.foodhyd.2020.106011>.

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